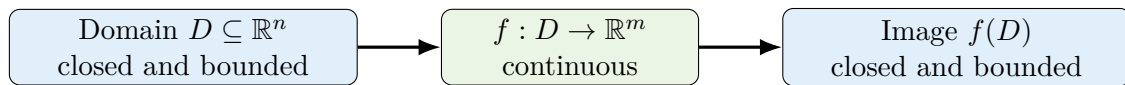


Open/Closed and Bounded/Unbounded Sets

Visual Examples and Their Interaction with Domains and Ranges

AS1109 Calculus and Differential Equations



In Euclidean space, *closed + bounded* means compact.

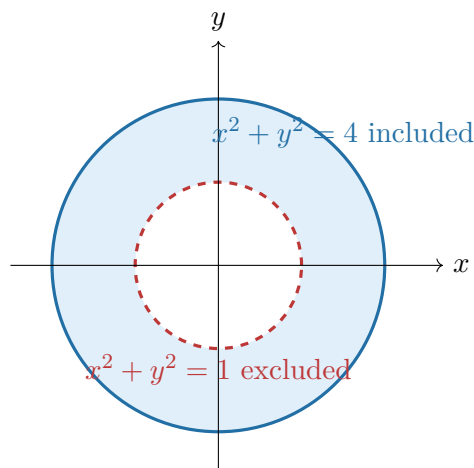
A continuous image of a compact set is compact. For scalar-valued functions, maxima and minima are attained.

1. A bounded set that is neither open nor closed

Consider

$$A = \{(x, y) : 1 < x^2 + y^2 \leq 4\}.$$

The outer circle is included, while the inner circle is excluded.



Solution. Since $x^2 + y^2 \leq 4$, every point lies within distance 2 of the origin, so A is bounded. It is not open because it contains points of the outer boundary. It is not closed because points of the inner circle are limit points of A but are missing.

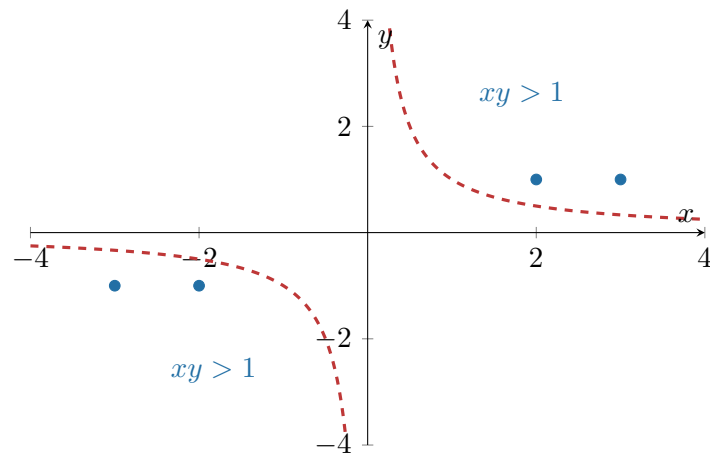
Classification: bounded; neither open nor closed

2. Open and unbounded: $xy > 1$

Let

$$B = \{(x, y) : xy > 1\}.$$

The boundary is the hyperbola $xy = 1$, and it is excluded.



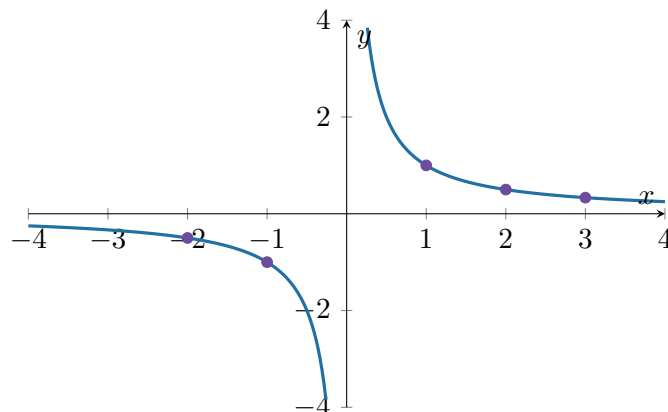
Solution. The map $(x, y) \mapsto xy$ is continuous, and $(1, \infty)$ is open, so B is open. It is unbounded because $(n, 1) \in B$ for every $n > 1$.

Classification: open and unbounded

3. Closed and unbounded: the hyperbola itself

Let

$$G = \{(x, y) : xy = 1\}.$$



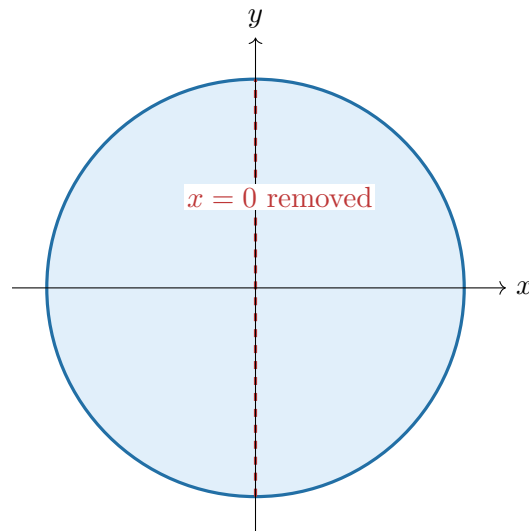
Solution. G is the inverse image of the closed set $\{1\}$ under the continuous function $f(x, y) = xy$, hence G is closed. It is unbounded since $(n, 1/n) \in G$ and its distance from the origin tends to infinity as $n \rightarrow \infty$.

Classification: closed and unbounded

4. A disk with a line removed

Consider

$$D = \{(x, y) : x^2 + y^2 \leq 9, x \neq 0\}.$$



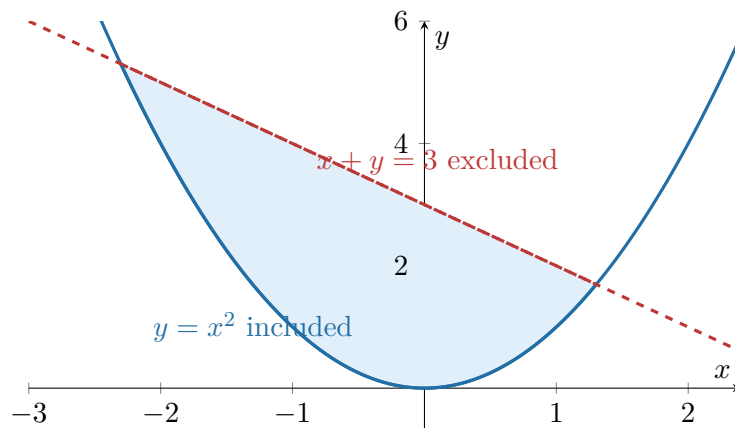
Solution. The disk makes the set bounded. It is not open because the circle $x^2 + y^2 = 9$ is included (except the two removed points on $x = 0$). It is not closed because, for example, $(1/n, 1) \in D$ for all sufficiently large n but $(1/n, 1) \rightarrow (0, 1) \notin D$.

Classification: bounded; neither open nor closed

5. Hidden boundedness: between a parabola and a line

Let

$$E = \{(x, y) : y \geq x^2, x + y < 3\}.$$



Solution. From $x^2 \leq y < 3 - x$ we obtain $x^2 + x - 3 < 0$. Hence

$$\frac{-1 - \sqrt{13}}{2} < x < \frac{-1 + \sqrt{13}}{2},$$

so x is bounded; then $x^2 \leq y < 3 - x$ also bounds y . The parabola boundary is included, so the set is not open. The line boundary is excluded but contains limit points, so the set is not closed.

Classification: bounded; neither open nor closed

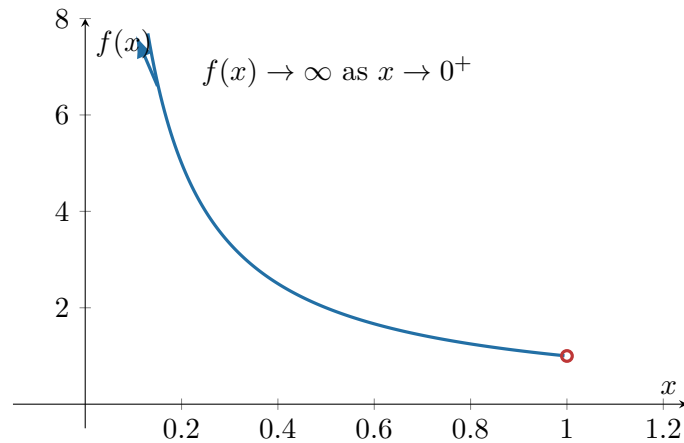
6. Bounded domain does not imply bounded range

Take

$$f(x) = \frac{1}{x}, \quad D = (0, 1).$$

The domain is bounded, but

$$f(D) = (1, \infty).$$



Reason. A bounded domain can approach a point where the function blows up. Compactness, not boundedness alone, is the crucial hypothesis.

$D = (0, 1) \text{ bounded,} \quad f(D) = (1, \infty) \text{ unbounded.}$

7. An unbounded domain can have a bounded range

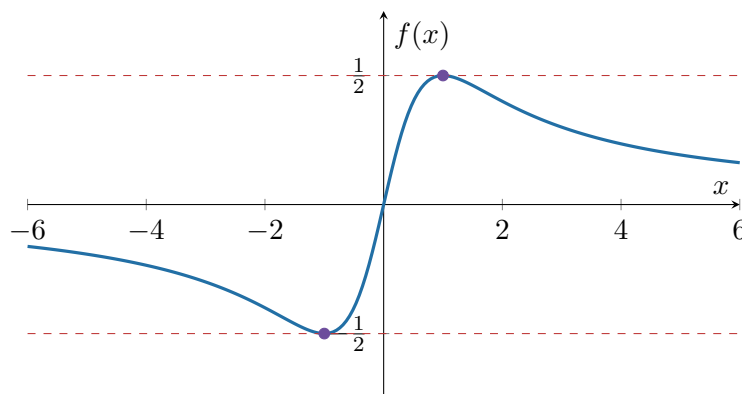
Let

$$f(x) = \frac{x}{1+x^2}, \quad D = \mathbb{R}.$$

Since

$$f'(x) = \frac{1-x^2}{(1+x^2)^2},$$

the critical points are $x = \pm 1$, with $f(1) = 1/2$ and $f(-1) = -1/2$.



Thus

$f(\mathbb{R}) = \left[-\frac{1}{2}, \frac{1}{2}\right],$

so an unbounded domain can have a bounded, closed range.

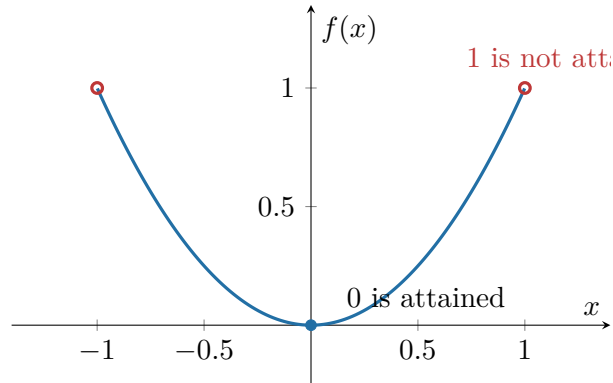
8. Open domain does not imply open range

Let

$$f(x) = x^2, \quad D = (-1, 1).$$

Then

$$f(D) = [0, 1).$$



Therefore the range is neither open nor closed.

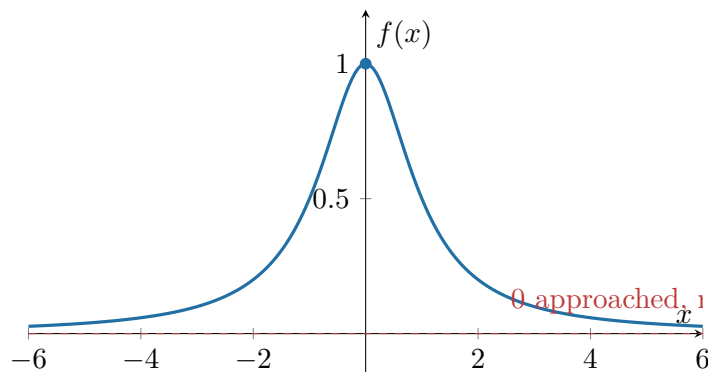
$$\boxed{(-1, 1) \xrightarrow{x^2} [0, 1).}$$

9. Closed domain does not imply closed range

Take

$$f(x) = \frac{1}{1 + x^2}, \quad D = \mathbb{R}.$$

Then $0 < f(x) \leq 1$, the value 1 is attained at $x = 0$, and $f(x) \rightarrow 0$ as $|x| \rightarrow \infty$ without ever reaching 0.



Hence

$$\boxed{f(\mathbb{R}) = (0, 1]},$$

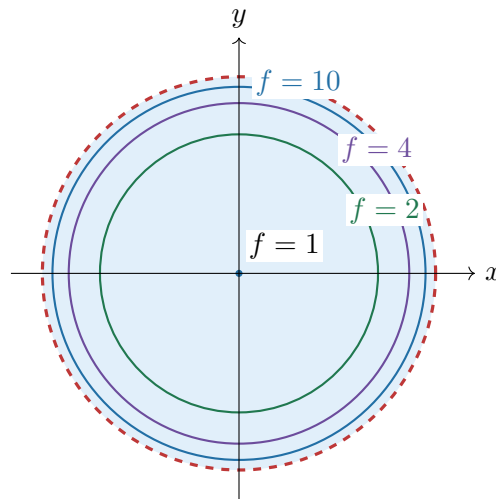
which is neither open nor closed.

10. A bounded open two-dimensional domain with unbounded values

Let

$$f(x, y) = \frac{1}{1 - x^2 - y^2}, \quad D = \{(x, y) : x^2 + y^2 < 1\}.$$

The domain is the bounded open unit disk, but $f(x, y) \rightarrow \infty$ as $x^2 + y^2 \rightarrow 1^-$.



boundary excluded; values blow up as the boundary is approached

Indeed, writing $r^2 = x^2 + y^2$ gives $0 \leq r^2 < 1$, so

$$f = \frac{1}{1 - r^2} \in [1, \infty).$$

Thus

D bounded and open, $f(D) = [1, \infty)$ unbounded.

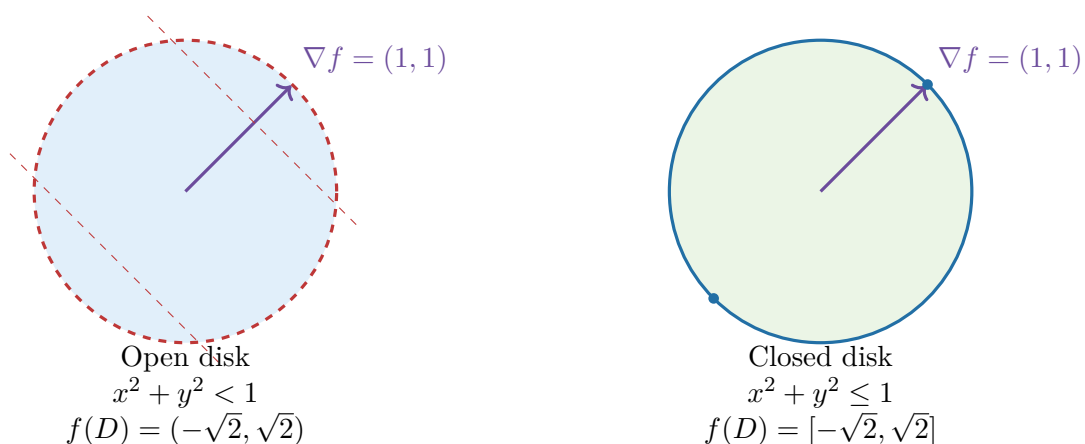
11. Open versus closed disk: the same function, different endpoint behaviour

Let

$$f(x, y) = x + y.$$

By Cauchy–Schwarz,

$$|x + y| \leq \sqrt{2} \sqrt{x^2 + y^2}.$$



The open disk permits values arbitrarily close to $\pm\sqrt{2}$ but does not attain them. The closed disk contains the two extremising boundary points

$$\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right), \quad \left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right),$$

so both extrema are attained.

12. Continuous image of a compact domain: a worked multivariable example

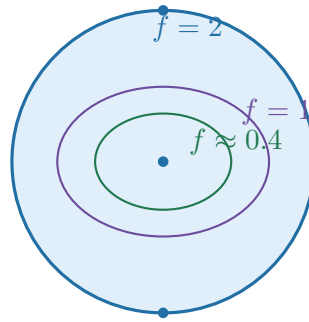
Let

$$f(x, y) = x^2 + 2y^2, \quad D = \{(x, y) : x^2 + y^2 \leq 1\}.$$

The domain is closed and bounded, and f is continuous. Also

$$0 \leq x^2 + 2y^2 \leq 2(x^2 + y^2) \leq 2.$$

The lower endpoint is attained at $(0, 0)$ and the upper endpoint at $(0, \pm 1)$.



$$f(D) = [0, 2]$$

Thus

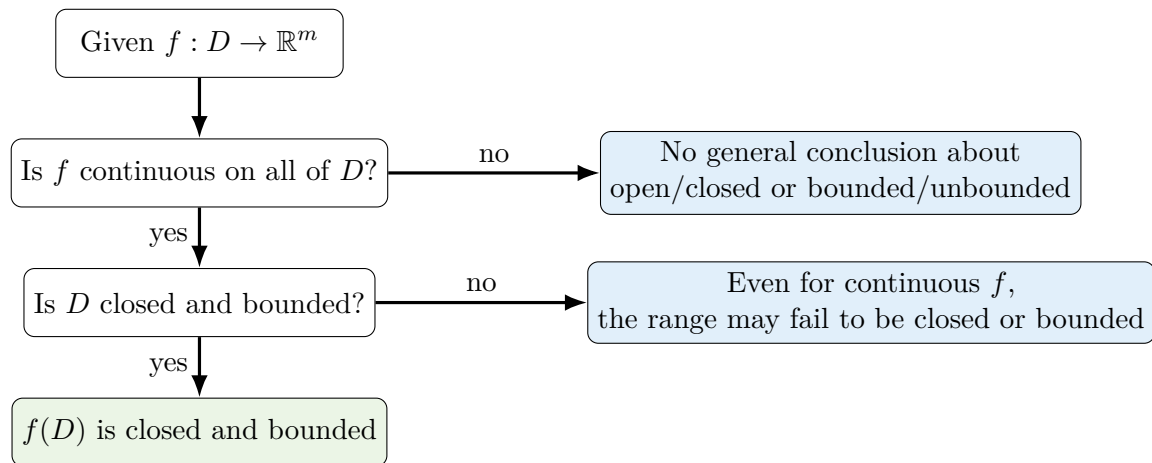
$$\boxed{f(D) = [0, 2]},$$

a closed and bounded range.

13. What properties are and are not preserved?

Claim	True?	Counterexample / reason
Bounded domain $\Rightarrow$ bounded range	No	$f(x) = 1/x$ on $(0, 1)$
Open domain $\Rightarrow$ open range	No	$f(x) = x^2$ on $(-1, 1)$ gives $[0, 1)$
Closed domain $\Rightarrow$ closed range	No	$f(x) = 1/(1 + x^2)$ on $\mathbb{R}$ gives $(0, 1]$
Unbounded domain $\Rightarrow$ unbounded range	No	$f(x) = x/(1 + x^2)$ gives $[-1/2, 1/2]$
Bounded range $\Rightarrow$ bounded domain	No	$\sin x$ maps $\mathbb{R}$ into $[-1, 1]$
Closed and bounded domain + continuous f $\Rightarrow$ closed and bounded range	Yes	Continuous image of a compact set is compact

14. A visual decision guide for classroom use



15. Quick board questions

For each statement, ask students to decide *true or false* and then demand either a theorem or a counterexample.

1. A bounded domain always gives a bounded range.
2. An open domain always gives an open range.
3. A closed domain always gives a closed range.
4. A continuous function on a closed and bounded domain has a bounded range.
5. A continuous real-valued function on a closed and bounded domain attains both a maximum and a minimum.
6. An unbounded domain can have a bounded range.

Answers: 1 False; 2 False; 3 False; 4 True; 5 True; 6 True.

Main takeaway: Continuity alone does not preserve “open”, “closed”, or “bounded” under images. In Euclidean space, the robust package is

continuous function + closed and bounded domain $\implies$ closed and bounded range.